

Genomska informatika - projektni zadaci

Prijava projekta se može obaviti slanjem e-maila na vladimir.kovacevic@etf.bg.ac.rs. Obavezno je navesti ime, prezime i broj indeksa. Projekat je moguće prijaviti i odbraniti u toku **cele** školske godine. Po pravilu, projekat se predaje i brani najkasnije **do termina ispita** u odgovarajućem ispitnom roku.

Rok za izradu projekta je **mesec dana** od dana prijave. U tom roku je potrebno poslati sve tražene rezultate na gore navedeni *e-mail*. Ukoliko studenti nisu u mogućnosti da završe projekat u predviđenom roku, treba o tome da obaveste predmetnog nastavnika radi eventualnog produžetka roka.

Projektni zadaci su predviđeni za timski rad. Po pravilu, timovi se sastoje od jednog ili dva studenta. Izuzetno, tim se može sastojati od tri studenta uz zadavanje dodatnih zahteva u okviru projekta.

Po pravilu, jedan projektni zadatak može raditi ograničen broj timova. Studenti mogu predložiti svoju temu projekta predmetnom nastavniku. Po potrebi, biće dodavane nove postavke projektnih zadataka tokom školske godine. Raspoloživost projektnih zadataka se može pratiti u tabeli ispod.

Upotreba AI agenata kao što su Claude code za razumevanje zadatka, predlog metoda i generisanje koda je dozvoljena, zbog čega projektni zahtevi neće biti definisani do najsitnijih detalja. Za sve što nije egzaktno definisano tekstem projekta uvesti logičnu pretpostavku. Slajdovi, prezentacija i komentari u programskom kodu poželjno je da budu na engleskom, ali je i srpski jezik u potpunosti prihvatljiv. **Poželjno je da se u toku razvoja projekta sav programski kod čuva na Github-u ili drugom sistemu za verzionisanje koda.** Budite kreativni!

	Studenti	Poeni
Zadatak 1		
Zadatak 1		

Zadatak 1

Single-Cell Analysis of Immune Response to Nanoplastic Particles

Nanoplastics that enter the bloodstream interact directly with immune cells, and particle size appears to shape the response. You will use single-cell RNA sequencing (scRNA-seq) to investigate how nanoplastics of different sizes affect human peripheral blood immune cells.

Dataset

Source: <https://zenodo.org/records/15866724> (DOI: 10.5281/zenodo.15866724)

Four samples from one donor, exposed to carboxylated polystyrene nanoparticles (PSNPs):

- **Sample 1**, 40 nm PSNPs
- **Sample 2**, 200 nm PSNPs
- **Sample 3**, 40 nm + 200 nm mixture
- **Sample 4**, control (no exposure)

Use the `.h5ad` files (AnnData format). The `*_CoDi_KLD.csv` files are used as a reference for potential comparison with the obtained results.

Tasks

1. **QC & preprocessing** — filter cells, normalise, select variable genes. Justify thresholds.
2. **Integration & clustering** - combine all four samples with a batch-correction method of your choice; compute UMAP and clusters.
3. **Cell type annotation** - assign immune cell types (T, B, NK, monocytes, etc.) using marker genes and the [Azimuth PBMC reference \(RDS\) dataset](#).
4. **Composition analysis** - compare cell type proportions across the four samples.
5. **Differential expression** - for each major cell type, compare each exposed sample against the control; run pathway enrichment (GO/Reactome/KEGG).
6. **Size-specific effects** - identify responses that are unique to 40 nm, unique to 200 nm, shared, or emerge only in the mixture. Interpret biologically.

Deliverables

- **Github Code repository** with an environment file and a README explaining how to reproduce results (20 points).
- Propose and implement 3-5 additional insights or analyses to the dataset (10 points)
- PowerPoint slides explaining all the results with visualisations (also saved on GitHub) (10 points)
- **Video presentation** (5-10 minutes) of the results uploaded to YouTube or other video platform (20 points)